# No Man's Sky Refining cheatsheet

self-multiplying loops · precious metals · best profit · how a refiner behaves

## ×6 EVERY 1m30s — the only 3 sixfold loops

| | | |
|---|---|---|
| **6 Condensed Carbon** | ← Condensed Carbon + 2 Oxygen | 240/h |
| **6 Ionised Cobalt** | ← Ionised Cobalt + 2 Oxygen | 240/h |
| **6 Chlorine** | ← Chlorine + 2 Oxygen | 240/h |

Feed the output back in. A Medium refiner (2 slots) turns 1 + 2 into 6 — the stack pays for itself and then some.

## SELF-MULTIPLYING LOOPS

every row: **Medium** refiner, **1m30s** a cycle, output feeds its own input. **out/h** = items made per machine-hour · **units** = net value gained per batch

| MAKE | FROM | OUT/H | UNITS |
|---|---|---|---|
| **6 Condensed Carbon** | Condensed Carbon + 2 Oxygen | 240 | 52 |
| **6 Ionised Cobalt** | Ionised Cobalt + 2 Oxygen | 240 | 742 |
| **6 Chlorine** | Chlorine + 2 Oxygen | 240 | 957 |
| **3 Sodium Nitrate** | Sodium Nitrate + Pugneum | 120 | 26 |
| **3 Condensed Carbon** | Faecium + Condensed Carbon | 120 | 18 |
| **3 Faecium** | Faecium + Oxygen | 120 | 26 |
| **2 Sodium Nitrate** | Sodium Nitrate + Oxygen | 80 | 48 |
| **2 Sodium Nitrate** | Nitrogen + Sodium Nitrate | 80 | 62 |
| **2 Sodium Nitrate** | Marrow Bulb + Sodium Nitrate | 80 | 41 |
| **2 Sodium Nitrate** | Kelp Sac + Sodium Nitrate | 80 | 41 |
| **2 Sodium Nitrate** | Faecium + Sodium Nitrate | 80 | 52 |
| **2 Sodium Nitrate** | Methane + Sodium Nitrate | 80 | 47 |
| **2 Ionised Cobalt** | Marrow Bulb + Ionised Cobalt | 80 | 121 |
| **2 Chlorine** | Kelp Sac + Chlorine | 80 | 164 |
| **2 Solanium** | Phosphorus + Solanium | 80 | 8 |
| **2 Cadmium** | Cadmium + Chromatic Metal | 80 | −5 |
| **2 Indium** | Indium + Chromatic Metal | 80 | 44 |
| **2 Emeril** | Emeril + Chromatic Metal | 80 | 26 |
| **2 Copper** | Copper + Chromatic Metal | 80 | −47 |
| **2 Star Bulb** | Paraffinium + Star Bulb | 80 | −30 |
| **2 Cactus Flesh** | Pyrite + Cactus Flesh | 80 | −34 |
| **2 Fungal Mould** | Ammonia + Fungal Mould | 80 | −46 |
| **2 Gamma Root** | Uranium + Gamma Root | 80 | −46 |
| **2 Frost Crystal** | Dioxite + Frost Crystal | 80 | −50 |
| **2 Quartzite** | Quartzite + Chromatic Metal | 80 | 44 |
| **4 Magnetised Ferrite** | 3 Magnetised Ferrite + Pugneum | 160 | −56 |

the negative rows still multiply — they just cost you value per cycle; run them for the **material**, not the profit.

## PRECIOUS-METAL ENGINE

Gold · Silver · Platinum · Chromatic Metal, by units per batch · the italic name is the operation to pick in the machine

**100 Gold** — 30 250
Living Pearl · 1m · P · 6 000/h · *Calcite Transmutation*

**250 Silver** — 21 500
Aronium · 5m · P · 3 000/h · *Extreme Alloy Separation*

**250 Silver** — 21 500
Dirty Bronze · 5m · P · 3 000/h · *Extreme Alloy Separation*

**250 Silver** — 21 500
Herox · 4m12s · P · 3 600/h · *Extreme Alloy Separation*

**125 Gold** — 19 125
Lemmium · 5m · P · 1 500/h · *Extreme Alloy Separation*

**125 Gold** — 19 125
Magno-Gold · 5m · P · 1 500/h · *Extreme Alloy Separation*

**125 Gold** — 19 125
Grantine · 5m · P · 1 500/h · *Extreme Alloy Separation*

**10 Gold** — 3 368
Ferrite Dust + Oxygen + Emeril · 20m · L · 30/h · *Chromatic Alchemy*

**30 Chromatic Metal** — 1 969
Silver + Gold + Indium · 20m · L · 90/h · *Chromatic Stellar Fusion*

**30 Chromatic Metal** — 1 969
Silver + Gold + Quartzite · 20m · L · 90/h · *Chromatic Stellar Fusion*

**20 Chromatic Metal** — 1 107
Silver + Gold + Emeril · 20m · L · 60/h · *Chromatic Stellar Fusion*

**2 Gold** — 538
Faecium + Pugneum · 1m30s · M · 80/h · *Pugneum Alchemy*

the alloys (Aronium, Herox, Lemmium…) are **stored money**: one unit unpacks into 125–250 metal in a portable refiner.

## BEST PROFIT PER BATCH

net units gained by one full run of the machine · one row per output

**Thermic Condensate** — 46 750
100 Sulphurine + 20 Carbon + 10 Salt · 1m30s · L · *High-Speed Sublimation*

**Enriched Carbon** — 46 750
100 Radon + 20 Carbon + 10 Salt · 1m30s · L · *High-Speed Sublimation*

**Nitrogen Salt** — 46 750
100 Nitrogen + 20 Carbon + 10 Salt · 1m30s · L · *High-Speed Sublimation*

**100 Gold** — 30 250
Living Pearl · 1m · P · *Calcite Transmutation*

**Warp Cell** — 23 740
25 Condensed Carbon + 10 Sodium + 250 Chromatic Metal · 10m · L · *Antimatter Bypass*

**250 Silver** — 21 500
Aronium · 5m · P · *Extreme Alloy Separation*

**Dirty Bronze** — 21 340
30 Pyrite + 120 Ferrite Dust + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Lemmium** — 21 340
30 Uranium + 120 Ferrite Dust + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Aronium** — 18 460
30 Paraffinium + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Herox** — 18 460
30 Ammonia + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Magno-Gold** — 18 460
30 Phosphorus + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Grantine** — 18 460
30 Dioxite + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Unstable Plasma** — 4 890
10 Oxygen + 20 Ferrite Dust + 20 Carbon · 1m30s · L · *Plasma Pressurisation*

**30 Chromatic Metal** — 1 969
Silver + Gold + Indium · 20m · L · *Chromatic Stellar Fusion*

## BEST UNITS PER MACHINE-HOUR

profit per batch × batches an hour — what one machine actually earns

**Nitrogen Salt** — 8 412 300
100 Nitrogen + 10 Condensed Carbon + 5 Chlorine · 20s · L · *High-Speed Sublimation*

**Thermic Condensate** — 1 870 000
100 Sulphurine + 20 Carbon + 10 Salt · 1m30s · L · *High-Speed Sublimation*

**Enriched Carbon** — 1 870 000
100 Radon + 20 Carbon + 10 Salt · 1m30s · L · *High-Speed Sublimation*

**100 Gold** — 1 815 000
Living Pearl · 1m · P · *Calcite Transmutation*

**Dirty Bronze** — 853 600
30 Pyrite + 120 Ferrite Dust + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Lemmium** — 853 600
30 Uranium + 120 Ferrite Dust + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Aronium** — 738 400
30 Paraffinium + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Herox** — 738 400
30 Ammonia + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Magno-Gold** — 738 400
30 Phosphorus + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**Grantine** — 738 400
30 Dioxite + 60 Cobalt + 20 Tritium · 1m30s · L · *Alloy Latticing*

**250 Silver** — 309 600
Herox · 4m12s · P · *Extreme Alloy Separation*

**Unstable Plasma** — 195 600
10 Oxygen + 20 Ferrite Dust + 20 Carbon · 1m30s · L · *Plasma Pressurisation*

**Warp Cell** — 142 440
25 Condensed Carbon + 10 Sodium + 250 Chromatic Metal · 10m · L · *Antimatter Bypass*

**Glass** — 43 200
40 Silicate Powder · 10s · P · *Silicate Forging*

## REFINER RULES

1. The output slot holds **exactly one stack** — the process auto-pauses the moment it fills. Empty it to keep going.
2. A lone leftover input is **consumed for nothing**, so you are never left holding a stack of 1.
3. Refiner contents **can vanish on quit**, and always vanish if you switch saves. Empty a refiner before you log off.
4. The **displayed refining time is frequently wrong** — every time on this sheet is read from the game files instead.
5. You can pull partial output or stop mid-run: **nothing is lost by interrupting** a batch.
6. Refined materials can be **refined again**, and higher tiers store more per slot — refine before you haul.

## MACHINES & SYMBOLS

| | SLOTS | MACHINE |
|---|---|---|
| **P** | 1 | Portable / Personal refiner |
| **M** | 2 | Medium refiner |
| **L** | 3 | Large refiner |
| **N** | 2 | Nutrient Processor (cooking) |

a **Medium** means two input slots, a **Large** three — the letter after an input list is the smallest machine that can run it.
**out/h** · **nan/h** items or nanites per machine-hour · **units** net value of one batch, green pays, red burns value · *italics* the operation to pick in the machine menu

# No Man's Sky Nanites cheatsheet



## NANITE CONVERTERS — EVERY SOURCE, BEST RATE FIRST

nanites are the second currency: **technology upgrades**, not ships and freighters. Nothing sells for nanites, so every row is really a price — the **sale value** figure is what the batch costs you in units, not a loss.
**nan** per batch · **nan/h** per machine-hour · **per item** nanites per item consumed · the coloured label is how hard the input is to come by (**easy ok work hard**). The **where** line is the part the game files never tell you.
**P** portable refiner (1 slot) · **M** medium (2) · **L** large (3) · *italics* = the operation to pick in the machine

**Hadal Core** — **50** nan · 3 000/h · 50/item
loot (medium) · Underwater — hives and the sea floor
1m · P · *Doom Cycling* · sale value −96 500

**Larval Core** — **50** nan · 3 000/h · 50/item
loot (medium) · Whispering Eggs, outside abandoned buildings
1m · P · *NOT RECOMMENDED* · sale value −64 000 · *Fastest nanites in the game per refiner, tied with Hadal Core.*

**Vile Spawn** — **50** nan · 2 000/h · 50/item
loot (medium) · Hives
1m30s · P · *Loosen Bonds* · sale value 984

**Flesh Rope** — **50** nan · 2 000/h · 50/item
loot (medium) · Hives, underwater, Whispering Eggs
1m30s · P · *Loosen Bonds* · sale value −15 016

**Hyaline Brain** — **230** nan · 259/h · 230/item
loot (hard) · Corrupted-planet creatures, while hunting Sentinel Interceptors
53m · P · *Sentience Extraction* · sale value −30 400 · *230 nanites in one inventory slot is unbeatable storage, but 53 min each: compact, not fast.*

**Salvaged Data** — **15** nan · 169/h · 15/item
loot (easy) · Buried beside Damaged Machinery — dig with the Terrain Manipulator
5m18s · P · *Nanite Extraction* · sale value −51 700 · *The Space Anomaly blueprints eat a few hundred; refine the rest at 15 nanites each.*

**Inverted Mirror** — **95** nan · 155/h · 95/item
loot (medium) · Byproduct of shooting Dissonant Resonators for Echo Locators
37m · P · *Purge Corruption* · sale value −15 100 · *Piles up unused during Echo Locator farming — refine the surplus.*

**25 Pugneum** — **1** nan · 120/h · 0.04/item
loot (easy) · Destroyed Sentinels
30s · P · *Reality Filtering* · sale value −3 430

**Radiant Shard** — **50** nan · 100/h · 50/item
loot (medium) · Hives, underwater, Whispering Eggs
30m · P · *Subjugation of Glass* · sale value −900

**Tainted Metal** — **2** nan · 80/h · 2/item
loot (medium) · Derelict freighters, Corrupted Sentinels, Autophage camps
1m30s · P · *Doom Cycling* · sale value −340 · *Highest volume, not highest per unit: 150-500 from one derelict run.*

**35 Platinum** — **1** nan · 60/h · 0.03/item
T1 raw · Asteroids — the rarer gold-tinted ones
1m · P · *Reality Warping* · sale value −17 655

**Faecium + Hexite** — **1** nan · 60/h · 0.5/item
loot (hard) · Guild Envoy reward
1m · M · *Alchemical Construction* · sale value −664

**15 Silver + 15 Gold + 25 Platinum** — **1** nan · 40/h · 0.02/item
T1 raw · Asteroids — shoot them in any system
1m30s · L · *Transmutation* · sale value −20 690

**5 Runaway Mould** — **1** nan · 24/h · 0.2/item
loot (easy) · Curious Deposits, which respawn every time you return to the planet
2m30s · P · *Recycle Waste Materials* · sale value −80 · *Gathering is trivial; at 5:1 and 150 s per nanite the refining is the whole cost.*

## ESSENTIALS — STOP MINING THESE

the cheapest route to what you would otherwise go and dig up, using only materials the source tiers below actually reach · **seed:** marks the one input you have to find yourself before the loop runs

| MAKE | FROM · OPERATION | TIME | M |
|---|---|---|---|
| **30 Chromatic Metal** | Silver + Gold + Quartzite · *Chromatic Stellar Fusion* · **seed:** refinable once you have Activated Quartzite | 20m | L |
| **5 Condensed Carbon** | 2 Carbon + 2 Oxygen · *Oxygenate Carbon* | 1m30s | M |
| **2 Sodium Nitrate** | Condensed Carbon + Sodium · *Catalyse Carbon* | 1m30s | M |
| **10 Magnetised Ferrite** | Platinum + Oxygen · *Mineral Alchemy* | 1m30s | M |
| **2 Pure Ferrite** | Magnetised Ferrite · *Demagnetise Metal* | 40s | P |
| **2 Ionised Cobalt** | Marrow Bulb + Oxygen · *Bonded Cobalt Extraction* | 1m30s | M |
| **Deuterium** | Di-hydrogen + Tritium · *Di-hydrogen Cycling* | 1m | M |
| **Starship Launch Fuel** | 10 Di-hydrogen + 20 Ferrite Dust + 5 Sodium Nitrate · *Di-hydrogen Capture* | 1m30s | L |
| **2 Pyrite** | Gold + Oxygen · *Mineral Alchemy* | 1m30s | M |
| **Salt** | Di-hydrogen + Oxygen · *Rapid Formation / Evaporation* | 20s | M |
| **Glass** | 40 Frost Crystal · *Polish Crystals* | 1m | P |
| **10 Oxygen** | Frost Crystal + Kelp Sac + Condensed Carbon · *Artificial Photosynthesis* | 15m | L |

## TIER 0 — FROM FIVE MATERIALS

a refiner plus Oxygen, Carbon, Ferrite Dust, Di-hydrogen, Sodium makes all 30 of these:

Ammonia
Basalt
Cactus Flesh
Chlorine
Cobalt
Condensed Carbon
Di-hydrogen Jelly
Dioxite
Faecium
Frost Crystal
Fungal Mould
Gamma Root
Glass
Ionised Cobalt
Life Support Gel
Magnetised Ferrite
Marrow Bulb
Mordite
Paraffinium
Phosphorus
Pure Ferrite
Pyrite
Rusted Metal
Salt
Sodium Nitrate
Solanium
Star Bulb
Starship Launch Fuel
Unstable Plasma
Uranium

## TIER 1 — ADD ASTEROID MINING

Silver, Gold, Platinum, Tritium off asteroids unlocks 9 more, nanites included:

Aronium
Deuterium
Dirty Bronze
Grantine
Herox
Ion Battery
Lemmium
Magno-Gold
Nanite Cluster

## MUST FIND — NO RECIPE MAKES THESE

25 items **no refiner recipe produces**: derelicts, dissonant worlds, abandoned buildings, sentinel pillars, the sea floor. Everything else on this sheet is downstream of them.

Activated Cadmium
Activated Copper
Activated Emeril
Activated Indium
Atlantideum
Crystal Sulphide
Cursed Dust
Cyto-Phosphate
Flesh Rope
Geodesite
Hadal Core
Hexite
Hyaline Brain
Hypnotic Eye
Inverted Mirror
Iridesite
Larval Core
Living Pearl
Polished Stone
Radiant Shard
Salvaged Data
Silicate Powder
Tainted Metal
Tritium
Vile Spawn

# No Man's Sky Kitchen cheatsheet

## COOK ONCE, EAT ONE, SELL THE REST

the dish you eat is the dish you sell — there is no "eat or sell?"

A Nutrient Processor batch takes seconds and comes out as both the buff and the payday: the same cake that tops your Shield Strength sells for six figures. Cook a stack, eat one, sell the rest.

## COOKING FOR MONEY — TOP DISHES BY VALUE

every row below: **5s** in the Processor, so units/hour = value × 720 — the best row is 70 272 000 an hour

| DISH | FROM · OPERATION | UNITS |
|---|---|---|
| **Ambrosial Curse** | Writhing, Roiling Batter + Syrupy Nectar · *Assemble Baked Product* | 97 600 |
| **Cream Fingers** | Sugar Dough + Cream · *Dough Augmentation* | 94 600 |
| **Bone Broth** | Softened Marrow + Sweetroot · *Combine And Reduce* | 92 200 |
| **Nectar Islands** | Extra-Fluffy Batter + Syrupy Nectar · *Assemble Baked Product* | 79 200 |
| **The Worst Stew** | Juicy Grub + Rancid Flesh · *Purge by Fire* | 78 800 |
| **Softened Marrow** | Any Edible Bone · *Stew Bones* | 77 400 |
| **Haunted Wafer** | Sugar Dough + Fiendish Roe · *Dough Augmentation* | 73 600 |
| **Cake of the Lost** | Writhing, Roiling Batter + Salty Custard · *Assemble Baked Product* | 71 200 |
| **Sea's Bounty** | Any Cooked Seafood ×3 · *Seafood Assembly* | 66 650 |
| **Horrifying, Gooey Delight** | Cake Batter + Monstrous Custard · *Assemble Baked Product* | 62 000 |
| **'Apple' Cake of Lost Souls** | Wailing Batter + Crab 'Apple' · *Assemble Baked Product* | 53 700 |
| **Fish Biscuit** | Sugar Dough + Salty Chunks · *Dough Augmentation* | 53 600 |
| **Soft Custard Fancy** | Extra-Fluffy Batter + Salty Custard · *Assemble Baked Product* | 52 800 |
| **Appalling Jam Sponge** | Wailing Batter + Grahj'am + Cream · *Assemble Baked Product* | 51 900 |

## COOKING FOR BUFFS — THE BEST DISHES FOR EVERY EFFECT

19 stats a meal can boost, each with the three best-paying dishes that boost it · **pot** is the game's own 0–1 rating of the buff (a couple go past 1) · **units** is what the dish sells for · **uplift** is what one batch gains over its ingredients · ties are broken by price, so the **first dish of each effect is both the strongest buff and the best sale**

| DISH | FROM | POT | UNITS | UPLIFT |
|---|---|---|---|---|
| **Sprint Distance** — crossing ground on foot · 20 dishes | | | | |
| Refined Flour | Heptaploid Wheat | 1.8 | 1 800 | +1 000 |
| The Spawning Tart | Pie Case + Monstrous Custard | 1 | 160 000 | +30 000 |
| Haunted Pie | Pie Case + Fiendish Roe | 1 | 52 000 | +20 600 |
| **Breathing Efficiency** — underwater exploration · 37 dishes | | | | |
| Deathly-Cold Ice Cream | Monstrous Custard + Frozen Tubers | 1 | 128 000 | +27 200 |
| Sea's Bounty | Any Cooked Seafood ×3 | 1 | 71 450 | +66 650 |
| Salty Juice | Cactus Flesh + Salt + Fire Water | 0.9 | 1 800 | −129 |
| **Cold Resistance** — frozen planets · 13 dishes | | | | |
| Iced Screams | Hypnotic Eye + Frozen Tubers | 1 | 90 000 | +29 200 |
| Caramel Ice Cream | Viscous Custard + Frozen Tubers + Crunchy Caramel | 1 | 56 000 | +17 800 |
| 'Apple' Ice Cream | Viscous Custard + Frozen Tubers + Crab 'Apple' | 1 | 56 000 | +20 000 |
| **Fauna Analysis Rewards** — scanning creatures for units · 22 dishes | | | | |
| Cake of the Lost | Writhing, Roiling Batter + Viscous Custard | 1 | 199 200 | +67 200 |
| Ambrosial Curse | Writhing, Roiling Batter + Syrupy Nectar | 1 | 199 200 | +97 600 |
| Gooey Screamer | Thick, Sweet Batter + Monstrous Custard | 1 | 167 200 | +19 200 |
| **Flora Analysis Rewards** — scanning plants for units · 17 dishes | | | | |
| Appalling Jam Sponge | Wailing Batter + Grahj'am + Cream | 1 | 180 700 | +51 900 |
| Primordial Sponge | Wailing Batter + Sweetened Mucous + Bone Cream | 1 | 180 700 | +19 700 |
| 'Apple' Cake of Lost Souls | Wailing Batter + Crab 'Apple' | 1 | 176 900 | +53 700 |
| **Hazard Protection Battery** — extreme planets, storms · 24 dishes | | | | |
| Baked Cheese Tart | Pie Case + Tangy Cheese | 1 | 62 000 | +23 000 |
| Cheesy Vegetable Pie | Pie Case + Tangy Cheese + Solartillo | 1 | 62 000 | +22 700 |
| Proto-Beignet | Proto-Oil + Dough + Processed Sugar | 0.8 | 72 000 | +19 200 |
| **Heat Resistance** — scorched planets · 23 dishes | | | | |
| Cake of Burning Dread | Wailing Batter + Ever-burning Jam + Cream | 1 | 176 900 | +48 100 |
| Burning Jam Surprise | Writhing, Roiling Batter + Ever-burning Jam + Cream | 1 | 149 400 | +40 600 |
| Startling Fancy | Proto-Batter + Ever-burning Jam + Cream | 1 | 78 000 | +21 200 |
| **Jetpack Tanks** — vertical caves, cliff bases · 22 dishes | | | | |
| Horrifying, Gooey Delight | Cake Batter + Monstrous Custard | 1 | 200 000 | +62 000 |
| Cake of Glass | Wailing Batter + Anomalous Jam + Cream | 1 | 179 600 | +48 800 |
| Unbound Monstrosity | Writhing, Roiling Batter + Anomalous Jam + Cream | 1 | 152 200 | +41 400 |
| **Life Support Tanks** — long surface treks · 15 dishes | | | | |
| Muculent Tart | Pie Case + Sweetened Mucous | 1 | 62 000 | +23 000 |
| Seeping Pie | Pie Case + Juicy Thorax | 1 | 62 000 | +30 600 |
| Honey Tart | Pie Case + Sticky 'Honey' | 1 | 48 000 | +14 800 |
| **Mineral Analysis Rewards** — scanning minerals for units · 24 dishes | | | | |
| Cake of Sin | Writhing, Roiling Batter + Crab 'Apple' | 1 | 149 400 | +46 200 |
| Gooey Custard Fancy | Thick, Sweet Batter + Viscous Custard | 1 | 97 200 | +17 200 |
| Sponge of Ambrosia | Syrupy Batter + Viscous Custard | 1 | 97 200 | +17 200 |
| **Mining Speed** — mining runs · 7 dishes | | | | |
| Gooey ProtoDoughnut | Lumpen Doughnut + Gooey ProtoButter | 1 | 120 000 | +36 000 |
| Custard Doughnut | Lumpen Doughnut + Viscous Custard | 1 | 105 000 | +25 000 |
| Honeybutter Doughnut | Lumpen Doughnut + Sweetened Butter | 1 | 105 000 | +31 000 |
| **Radiation Resistance** — irradiated planets · 4 dishes | | | | |
| Glowing Pie | Pie Case + Fireberry | 1 | 40 000 | +9 200 |
| Grahj'am | Grahberry + Processed Sugar | 0.8 | 3 400 | +1 000 |
| Wriggling Tack | Sugar Dough + Nightmare Sausage | 0.66 | 55 000 | +34 000 |
| **Recharge Time** — keeping tech topped up · 6 dishes | | | | |
| Monstrous Doughnut | Lumpen Doughnut + Monstrous Custard | 1 | 180 000 | +32 000 |
| Pollen Puffball | Lumpen Doughnut + Butter Syrup | 1 | 120 000 | +40 000 |
| Custard Fancy | Cake Batter + Viscous Custard | 1 | 85 100 | +15 100 |
| **Resources Mined** — mining runs - more per rock · 20 dishes | | | | |
| The Stellarator | Lumpen Doughnut + Stellar Custard | 1 | 100 000 | +24 000 |
| Questionably Sweet Cake | Cake Batter + Sticky 'Honey' | 1 | 73 600 | +32 400 |
| Syrup-Drenched Delight | Thick, Sweet Batter + Syrupy Nectar | 1 | 73 600 | +24 000 |
| **Shield Strength** — combat, sentinels · 16 dishes | | | | |
| Cake of Eternal Sleep | Wailing Batter + Syrupy Nectar | 1 | 167 200 | +45 600 |
| Volatile Chocolate Fancy | Writhing, Roiling Batter + Bittersweet Cocoa | 1 | 123 700 | +21 900 |
| Cream Curiosity | Proto-Batter + Cream | 1 | 64 900 | +11 500 |
| **Solar Panel Power** — exocraft and base power · 15 dishes | | | | |
| Caramelised Nightmare | Writhing, Roiling Batter + Crunchy Caramel | 1 | 128 000 | +22 600 |
| Jam Doughnut | Lumpen Doughnut + Ever-burning Jam | 1 | 70 000 | +18 600 |
| Honied Proto-Cake | Proto-Batter + Synthetic Honey | 1 | 62 200 | +10 800 |
| **Sprint Recovery Time** — crossing ground on foot · 30 dishes | | | | |
| Monstrous Custard | Larval Core + Cream + Processed Sugar | 1 | 100 000 | +28 000 |
| Custard Tart | Pie Case + Viscous Custard | 1 | 78 000 | +16 000 |
| Creamy Treat | Pie Case + Cream | 1 | 62 000 | +26 600 |
| **Toxic Resistance** — toxic planets · 12 dishes | | | | |
| Briney Delight | Cake Batter + Salty Custard | 1 | 80 200 | +14 200 |
| Mucal Doughnut | Lumpen Doughnut + Sweetened Mucous | 1 | 70 000 | +13 000 |
| Slime Pop | Very Thick Custard + Frozen Tubers + Sweetened Mucous | 1 | 44 000 | +2 200 |
| **Words Translated** — learning alien languages fast · 4 dishes | | | | |
| Horrifying Mush | Hypnotic Eye | 1 | 90 000 | +30 000 |
| Abyssal Stew | Hypnotic Eye ×2 | 1 | 90 000 | −30 000 |
| Baked Anomaly | Sugar Dough + Anomalous Jam | 0.66 | 40 000 | +29 600 |

# No Man's Sky Multi-Tools cheatsheet

types · weapons · the free S-class Sentinel tool · supercharged slots · upgrade order

**THE TOOL IS THE BUILD** Sentinel and Voltaic Staff tie for the best damage in the game at +50%; Atlantid owns mining at +50%; Experimental and Royal have a fixed +100% scan. Pistols have no damage bonus at any class.

8 types · 7 weapons · the one page not generated from the recipe tables — it comes from the written guides.

## TOOL TYPES — WHAT THE CLASS RANGE ACTUALLY BUYS YOU

ranges run C-class to S-class · **green** = best in column · **red** = no bonus at all

| TYPE | DAMAGE | MINING | SCAN | SLOTS | |
|---|---|---|---|---|---|
| **Pistol** | +0% | +20-35% | +40-50% | 9-18 | Your starter. Literally no damage bonus at any class - that is why it feels the way it does. |
| **Rifle** | +15-20% | +0% | +15-20% | 13-30 | No mining bonus at all. |
| **Experimental** | +15-25% | +25-30% | +100% fixed | 9-30 | Scan is always exactly +100%, whatever the class. |
| **Alien** | +25-35% | +15-20% | +50-60% | 9-30 | The best scan outside the two fixed-100% types. |
| **Royal** | +15-25% | +25-30% | +100% fixed | always 20 | Only type with a fixed slot count. |
| **Sentinel** | +25-50% | +10-15% | +40-50% | 17-30 | Best damage in the game, tied with the Voltaic Staff. Carries the Hijacker Laser; only two design variants. |
| **Atlantid** | +10-20% | +35-50% | +40-50% | 17-30 | Best mining in the game. The Runic Lens triples mining beam efficiency and grants temporary invisibility. |
| **Voltaic Staff** | +25-50% | +10-15% | +40-50% | 17-30 | Ties the Sentinel for best damage. |

## GETTING THE RARE TYPES

**Sentinel** free Harmonic Camp on a dissonant planet
Solve the sealed-cabinet puzzle at the camp terminal. Best tool available early.

**Royal** free Sentinel Pillar terminals
Always spawns with 20 slots.

**Atlantid** free Korvax Monolith Discordant Interfaces
Charge with Atlantideum. The interfaces ignore you until the mission They Who Returned is complete.

**Voltaic Staff** Void Motes Autophage Synthesis Terminals
Buy the parts and assemble. The same head at the same terminal always produces the same class and colour - to reroll, assemble at a different terminal.

**Alien / Experimental** units Space station cabinets in wealthy systems
Wealthy systems have the highest S-class chance, and every station has two cabinets. Use the rotation trick.

## THE FREE S-CLASS SENTINEL TOOL, STEP BY STEP

1. Go to the camp's terminal, not the container.
2. Choose Scan Memory Registers - you get three addition problems, e.g. 4+11, 2+7, 3+1.
3. Choose Input Override Glyphs and enter the three answers (15, 9, 4). The glyphs are numbered.
4. Order does not matter - only the three correct numbers.
5. Deactivate the multi-tool seal, then take the tool from the container.

**Once it is yours**

- It arrives with broken nodes: repair with Radiant Shards and Atlantideum, the purple crystals on the same planet. Radiant Shards also refine to 50 nanites each.
- At the 6 active tool cap you must exchange, not acquire - decide which one dies before walking in.
- Strip the old tool first: Y (Xbox) / Triangle (PlayStation) on installed tech picks it back up, Terrain Manipulator included. You lose nothing by swapping.
- The same dissonance spikes also mark crashed Sentinel ships, so stack the trip.

## FINDING A SPECIFIC TOOL

**The cabinet rotation trick**

1. Visit a planet in the system.
2. Save and reload on the planet's surface.
3. Return to the space station - both cabinets have changed.

One cabinet pulls from the planetary pool, the other from the station's own rotation, which scales with the planet count. A six-planet system can offer up to 12 designs at one station.

**Hunting a specific tool from a location guide**

1. Get to the right galaxy (for Eissentam: featured bases at the Anomaly, a hub player, or the Artemis questline's green choice).
2. Turn multiplayer OFF before you portal - the tool instance is shared, and if someone else grabs it first it is gone for everyone.
3. Dial the portal address.
4. Look for a communication station marking the spot, or navigate to the published coordinates.
5. At the settlement, get out of your ship and reload the autosave that just triggered - the tool appears in the cabinet after the reload.
6. You still pay for it. The worked example cost 7.5 million units.

## WEAPONS — MEASURED BEST HIT

the last two are secondaries — no damage figure; they back up a primary

**Scatter Blaster** shotgun — **2,170**
+ Excellent damage, one-shots some enemies; the pick for derelict freighters
– Almost useless at range, hard to control held down

**Neutron Cannon** charge cannon — **1,600**
+ Area of effect, hits several enemies, charges with Carbon rather than ammo
– Must hold to charge for real damage

**Blaze Javelin** sniper — **1,184 x2**
+ 1-2 second stun, charges with Carbon
– Slow between shots, poor close range against groups

**Bolt Caster** burst fire — **657**
+ A great all-rounder, good at all ranges, excellent reload. The only weapon Sentinel and Autophage upgrades work with
– Lowest damage per shot

**Pulse Splitter** full auto — **284**
+ Fire rate shreds, bonus fire damage
– Chews through ammo, kicks on trigger pull

**Paralysis Mortar** secondary - stun — —
+ Strong against big enemies like Sentinel Walkers; its one upgrade also works with the Blaze Javelin. Reloads with Unstable Plasma
– Only one upgrade helps it, and it takes no adjacency bonus

**Plasma Launcher** secondary - explosive — —
+ Great against groups, and it does take adjacency upgrades
– The explosion is timed, not on impact - it bounces back at you. Mid-to-long range only

## RULES OF THUMB

**Damage Potential is not damage output** — It rolls up the installed weapons and damage tech, including the mining beam, and you can only fire one weapon at a time. There is no DPS readout anywhere in the game; the only real measurement is shooting something and reading the numbers. It is still the best measuring stick you have - strip the loadout to one thing and add pieces one at a time to watch the delta.

**Adjacency runs horizontally and vertically, never diagonally** — Group same-weapon-type upgrades together; a coloured border means the bonus is live. The bonus takes the class of the lowest member of the pair, so an A next to an S gives an A-class bonus. Some upgrades give no adjacency at all - no border ever appears, so stop trying to fit them into the block.

**Supercharged slots want to be a 2x2 block** — Four in a square all feed each other because adjacency is horizontal and vertical only; four in a line or scattered do not. This is fixed at spawn and is the single thing that separates a great tool from an average one - it is what to look for when hunting.

**Supercharge the weapon itself** — The weapon shows no visible stats, so the slot looks wasted. Swapping the weapon out for an A-class upgrade dropped a loadout from 24,000 to 14,000. Worth testing yourself: this rests on the Damage Potential number, and part of the community argues the opposite.

**Put the Optical Drill in a supercharged slot** — It boosts efficiency by 88%, nearly doubling mining output - one of the strongest early upgrades in the game.

**Add the ordinary Mining Laser to a Sentinel tool** — Alongside the Sentinel's own laser it takes Damage Potential from 469 to 1,164 with no weapons installed, slows overheating and massively boosts mining.

**Scanner Range above ~700 is wasted** — The game does not display critical survival resources beyond roughly a 600-unit radius even with Scanner Range over 2,000.

**You do not need X-class** — X-class rolls come from pirate/outlaw station vendors and are a lottery - they can come out worse than C. The right tool with a good supercharged arrangement and the best S-class upgrades handles anything the game throws at you.

**Do not rush slot expansion** — Each slot costs progressively more, eventually 600 million units, and filling one tool with units alone runs past 10 billion. Saved Multi-tool Expansion Slots are worth hundreds of millions each. Upgrade class with nanites first: max slots are C 21, B 30, A 45, S 60.

**There is no need to chase a natural S-class** — Any tool upgrades to S at the station Upgrade Station with nanites. Pick a design you like and upgrade it yourself.

**You can hold 6 active and 18 archived** — The archive is reached through a Weapon Rack - buildable on a base, a corvette or a freighter. Salvage unwanted tools at the Anomaly's Decommissioning Terminal, which can even return Expansion Slots.

## UPGRADES — WHERE THEY COME FROM, WHAT TO FIT FIRST

**Craftable** — Blueprints learned at the Anomaly · Max 3-4 per weapon; some add utility rather than damage
**Purchased** — Multi-tool vendor at any space station · Max 3 per weapon, C to B to A to S
**X-class** — Space stations in pirate/outlaw systems · Random roll - can beat S-class or come out worse than C
**Sentinel / Autophage** — Extract Components on Salvaged Glass from dead Sentinels · On the multi-tool side these only work with the Bolt Caster

1. Mining beam first: Advanced Mining Laser (unlocks advanced deposits and Radiant Shards on dissonant worlds), then Optical Drill.
2. Scanner upgrades - they directly improve exploration income.
3. One main weapon with proper upgrades, for when fighting is unavoidable.
4. Stick to one primary and one secondary; spreading upgrades across weapons wastes slots.

# No Man's Sky Portals cheatsheet



**TWO NAMES FOR ONE PLACE** The glyphs you dial count from the **centre** of the galaxy; the address a signal booster shows counts from a **corner**. Same voxel, unrecognisable numbers — which is the whole reason a converter exists.

16 glyphs · 15 addresses (0 found here, 15 shared) · press `P` in the app to convert one

## THE 16 GLYPHS — THE DIAL IS IN THIS ORDER

the hex digit **is** the position, so `A` is the eleventh symbol every time · you need all 16 unlocked, not twelve

| # | HEX | GLYPH | # | HEX | GLYPH |
|---|---|---|---|---|---|
| 1 | 0 | sunset | 9 | 8 | dragonfly |
| 2 | 1 | bird | 10 | 9 | galaxy |
| 3 | 2 | face | 11 | A | voxel |
| 4 | 3 | diplo | 12 | B | fish |
| 5 | 4 | eclipse | 13 | C | tent |
| 6 | 5 | balloon | 14 | D | rocket |
| 7 | 6 | boat | 15 | E | tree |
| 8 | 7 | bug | 16 | F | atlas |

## READING AN ADDRESS

`p SSS YY ZZZ XXX`

| | |
|---|---|
| p | planet index in the system — a galactic address does **not** carry it, so you pick one |
| SSS | star system index |
| YY | voxel height |
| ZZZ | voxel width |
| XXX | voxel length |

**Worked example**

dial 

glyphs `1 1AC 00 8F4 025`
galactic `0824:007F:00F3:01AC`
voxel `X +37 Y +0 Z -1804`

## CONVERTING BY HAND

all arithmetic is hex; the system index crosses over unchanged

| | GLYPHS → GALACTIC | GALACTIC → GLYPHS |
|---|---|---|
| X | `(X + 2047) % 4096` | `(X + 2049) % 4096` |
| Y | `(Y + 127) % 256` | `(Y + 129) % 256` |
| Z | `(Z + 2047) % 4096` | `(Z + 2049) % 4096` |

The anchor: glyphs `000000000000` are the centre of the galaxy, and galactic coordinates call the same place `07FF:007F:07FF`. Get a constant wrong and that moves.

## BEFORE YOU DIAL

**The galaxy is not in the address.** — the same twelve glyphs mean a different star in each of the 256 galaxies, so an address shared without its galaxy is ambiguous. Everything below is **Euclid**, galaxy 1.
**Trust the address, not the label.** — the location is seeded to the coordinate and generally survives a patch. Class, slot count and headline damage are submitter metadata and rot.
**Reload location is part of the address.** — a design is seeded by where you last loaded, so one monolith yields several tools. An entry naming a reload planet is telling you which roll it saw.
**Activating a portal is not knowing glyphs.** — charging the four pillars only powers the machine; the glyphs are what lets you type a destination. **Artemis → The Purge** grants one per warp.

## FOUND HERE — MY OWN ADDRESSES

nothing yet. In the app: paste or decode an address, press `S`, set *Found* to `mine` — it is written to `data/locations.json` and appears here on the next `mise run pdf`.

## SHARED — READ OFF THE CATALOGUES

Portal Repository, read 2026-09-07, via the Suika Box guides 21.multitool-damage.md and 14.starships-ranked.md. The verified field is the address; class and slot claims are submitter metadata and rot.

**Nazen 60/Z6** — multi-tool — `6046FFAE4450`
Atlantid - the rare one that is actually in Euclid · Korvax Monolith. Needs They Who Returned finished and Atlantideum in hand; bookmark it
`0C4F:007E:02E3:0046` · Portal Repository / guide 21 · 2026-09-07

**Invernia Prime** — multi-tool — `21E9FDBF25B5`
Royal - always 20 slots, +100% scan · Sentinel Pillar at -10.76 / +42.39. Volcanic, up to 300F - bring fire protection
`0DB4:007C:03F1:01E9` · Portal Repository / guide 21 · 2026-09-07

**Cruxlag Scrapyard** — multi-tool — `4012016B5DE8`
Royal · +30.41 / -57.38, base built at the portal. Reload on Otsub for the White/Gold variant
`05E7:0080:0EB4:0012` · Portal Repository / guide 21 · 2026-09-07

**Udley Sigma** — multi-tool — `518C07001FF4`
Rifle, DP 3541.8 / 24 slots - or a pistol at SR 591.7 · Two tools from one spot via autosave reload
`07F3:0086:0800:018C` · Portal Repository / guide 21 · 2026-09-07

**Iogerto Nukoka** — multi-tool — `120072AC4970`
24-slot rifle · +8.19 / +26.56
`016F:00F1:02C3:0200` · Portal Repository / guide 21 · 2026-09-07

**Hamonkle VI** — multi-tool — `11AC008F4025`
Grey/white rifle - and S-class Solar starships spawn in-system · Minor settlement. Stack the trip: tool and ship in one visit
`0824:007F:00F3:01AC` · Portal Repository / guide 21 · 2026-09-07

**Miamidl Prime** — multi-tool — `2128F4548C41`
Alien pistol · Submitted twice independently (PS4 and Switch), so the location is corroborated
`0440:0073:0D47:0128` · Portal Repository / guide 21 · 2026-09-07

**Aukeeves** — starship — `101508ABDBA2`
Blue Exotic, crashed - free claim, no units · Next to a player base, cold biome
`03A1:0087:02BC:0015` · Portal Repository / guide 14 · 2026-09-07

**UptonicaXIX** — starship — `105EFD9D4303`
Bubble Guppy Exotic - and an S-class multi-tool in-system · Ship at -59.25 / -71.99 with a repair station built around it; gun at +75.33 / -80.26
`0B02:007C:01D3:005E` · Portal Repository / guide 14 · 2026-09-07

**Konslis** — starship — `1006FD532B7E`
White and silver Exotic. Layout published: 3 supercharged mid 3x10 field, 1 bottom-left · By the seed rule that layout is exactly what you will get
`037D:007C:0D31:0006` · Portal Repository / guide 14 · 2026-09-07

**Dudleyt** — starship — `314802D019A6`
Red Guppy Exotic · Spawns at the space station
`01A5:0081:0500:0148` · Portal Repository / guide 14 · 2026-09-07

**Nadalmet** — starship — `1419056B91B9`
Ball/blob Exotic · Random spawn at the space station
`09B8:0084:0EB8:0419` · Portal Repository / guide 14 · 2026-09-07

**Tayurrhode** — starship — `204CFDC295C8`
Navy Exotic · Submitted on Survival
`0DC7:007C:0428:004C` · Portal Repository / guide 14 · 2026-09-07

**Icy tempest system** — starship — `121205370A0E`
Exotic in the first wave at the space station · System name not recorded by the submitter
`020D:0084:0B6F:0212` · Portal Repository / guide 14 · 2026-09-07

**Unnamed Exotic system** — starship — `20C17F4EF478`
Exotic · As soon as you warp go to the space station, do not reload - the first set of ships in is it
`0C77:00FE:0CEE:00C1` · Portal Repository / guide 14 · 2026-09-07